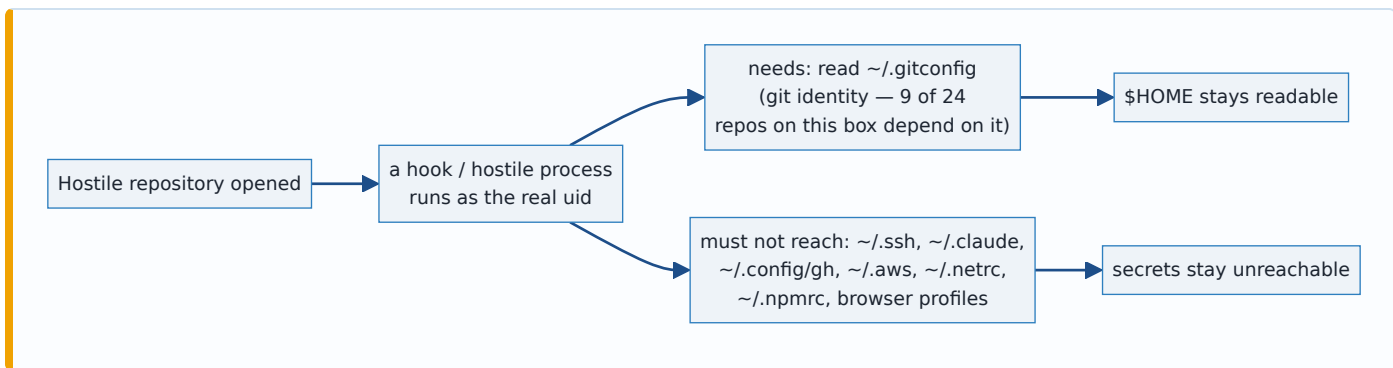


# 0027 — Secrets inside a granted `$HOME` are withheld by `enumerate-and-skip`, not a deny rule

- **Status:** Accepted
- **Date:** 2026-07-29
- **Milestone / issue:** M1.13b — the Git-process sandbox (#66). Corrects Global Constraint 9 of the round-4-settled plan ( `docs/superpowers/plans/2026-07-28-m1.13b-sandbox.md` ), ahead of Task 3 (the shim's Landlock enumeration, not yet built as of this ADR — see Consequences).
- **Supersedes:** Global Constraint 9 of the M1.13b plan, which specified the secret-withholding mechanism as *"a `path_beneath` rule with `allowed_access = 0`"*. That mechanism does not exist on this kernel — see Alternatives Considered. **Amends:** nothing directly. D5 Option B itself (read-allow `$HOME`, write-deny `$HOME`, plus an explicit secret list) is not re-litigated — only its enforcement mechanism is corrected.
- **Related:** [0025](#) (the same discipline — say plainly what is and is not true of the code — applied there to a half-shipped feature, applied here to a plan-level claim that turned out to be physically impossible); `docs/superpowers/specs/2026-07-28-m1.13-round4-verdict.md` (the settled design this plan implements); the M1.13b plan's own Task 18, which names its eventual whole-sandbox ADR `docs/adr/0026-git-process-sandbox.md` — that number is **already taken** by `0026-shell-mode-foundation.md` (confirmed by independent audit), so this ADR and any future whole-sandbox ADR both land above 0026, not at it. A reader who lands here from the plan's own text should not go looking for an "ADR 0026" that means this.

## Context

A hostile repository can reach code execution as the real uid through its own hooks ( `.git/hooks/*` ). The sandbox's read side must let `git` read the operator's real `$HOME` — nine of the twenty-four repositories inventoried on this box carry no repo-local `user.name / user.email` and depend on `~/.gitconfig` for identity, so a `$HOME` that isn't readable breaks ordinary commits, not just an edge case — while a fixed set of secret locations inside that same `$HOME` ( `~/.ssh`, `~/.claude`, `~/.config/gh`, `~/.aws`, `~/.netrc`, `~/.npmrc`, the Chrome/Chromium and Firefox profile directories) must stay unreachable to that same process.



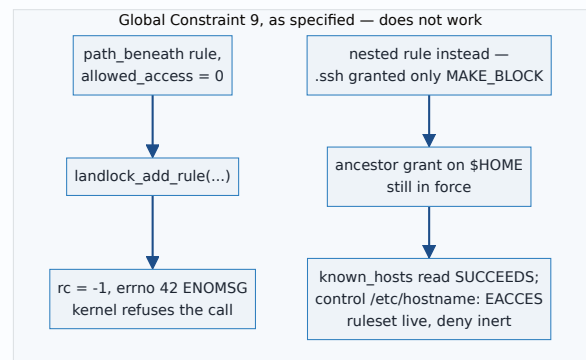
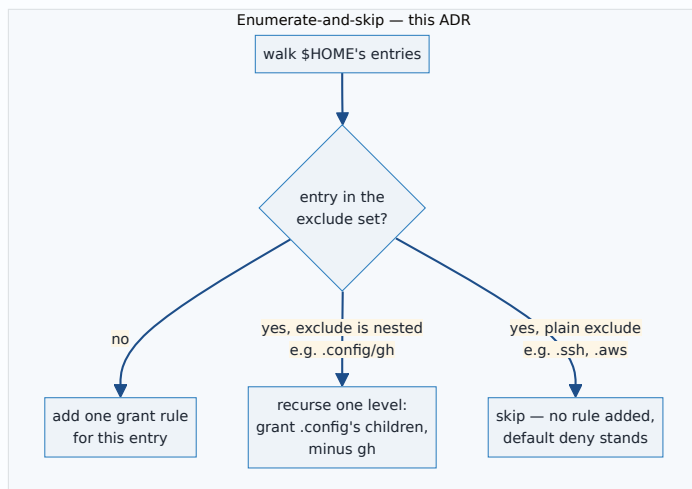
The M1.13b plan's Global Constraint 9 (round 4, "D5 Option B") specified the mechanism as a Landlock deny rule for the excluded paths. It does not exist:

1. **The kernel rejects it.** A `path_beneath` rule built with `allowed_access = 0` — the literal mechanism the plan named — is refused by `landlock_add_rule: measured rc = -1, errno = 42 (ENOMSG)`. The identical call with `READ_FILE` in place of `0` returns `0`. Since a non-zero return becomes `Err` and the shim's `restrict()` path uses `?`, shipping that mechanism as written would have aborted the shim on every launch on any host where `~/.ssh` exists — no sandboxed `git` process would ever run.
2. **The obvious alternative is inert.** A nested *lower-privilege* rule does **not** revoke rights an ancestor rule already granted. Measured: with `$HOME` granted `EXECUTE|READ_FILE|READ_DIR` and `$HOME/.ssh` separately granted only `MAKE_BLOCK`, reading `$HOME/.ssh/known_hosts` after `restrict_self()` returned `OK` **succeeded**, while the control `/etc/hostname` (no rule at all) correctly returned `EACCES` — proving the ruleset was live and the "deny" was simply inert, not that Landlock was disabled. Adding the nested rule before or after the parent grant changed nothing.

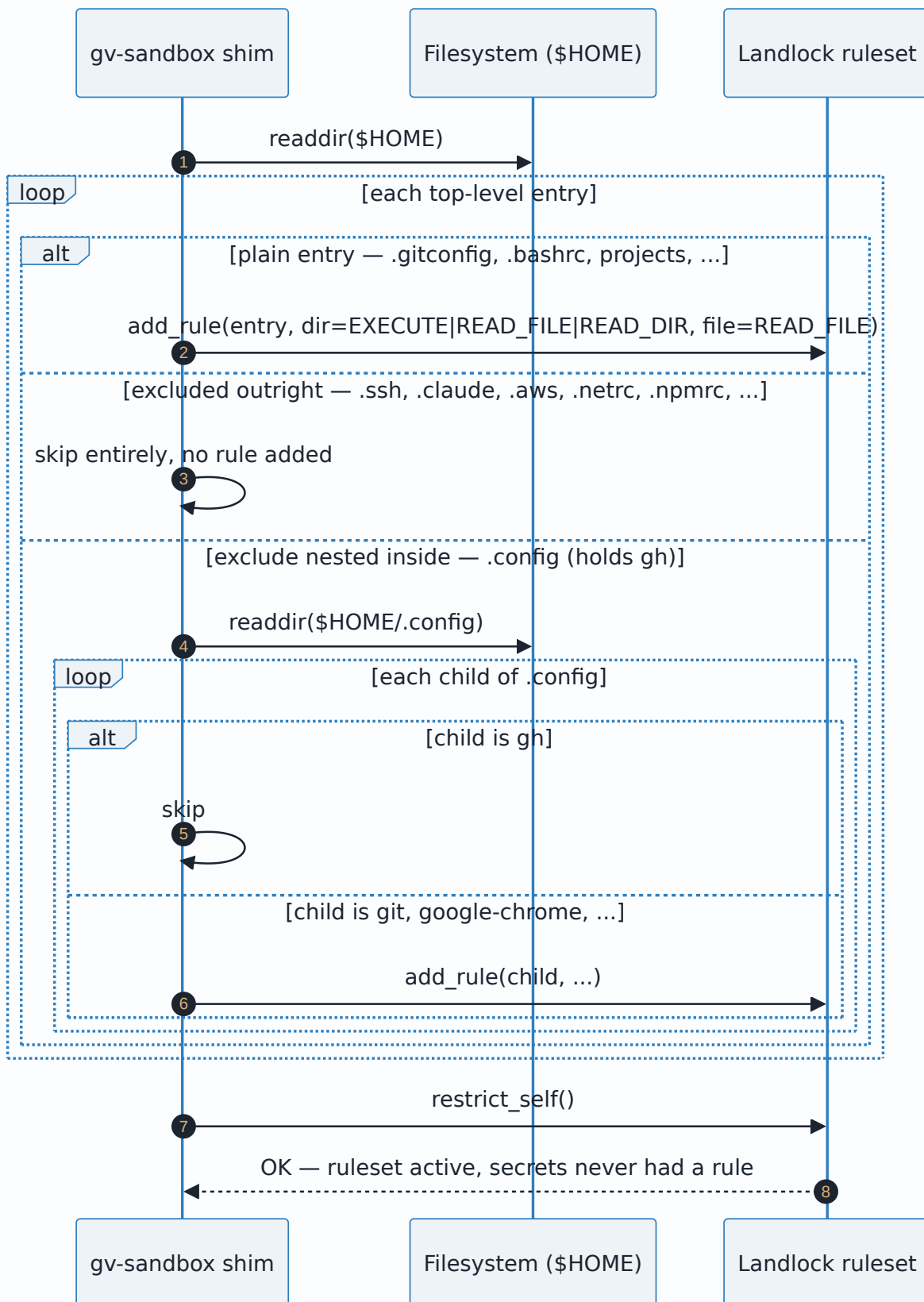
Landlock is deny-by-default. There is no rule shape that subtracts from an already-granted tree — denial can only be expressed by **never granting** the excluded path in the first place.

## Decision

Express denial as **not granting**. At policy-build time, enumerate the entries of every tree the policy intends to grant ( `$HOME` , foremost), and add one Landlock rule per entry — never one rule for the whole tree — skipping any entry that appears in the exclude set. Where an exclude sits *inside* a directory that is otherwise granted ( `.config/gh` , since `~/.config` itself must stay readable for `~/.config/git/ignore` ), recurse one level: grant that directory's children individually, minus the excluded one.



## Walking the tree

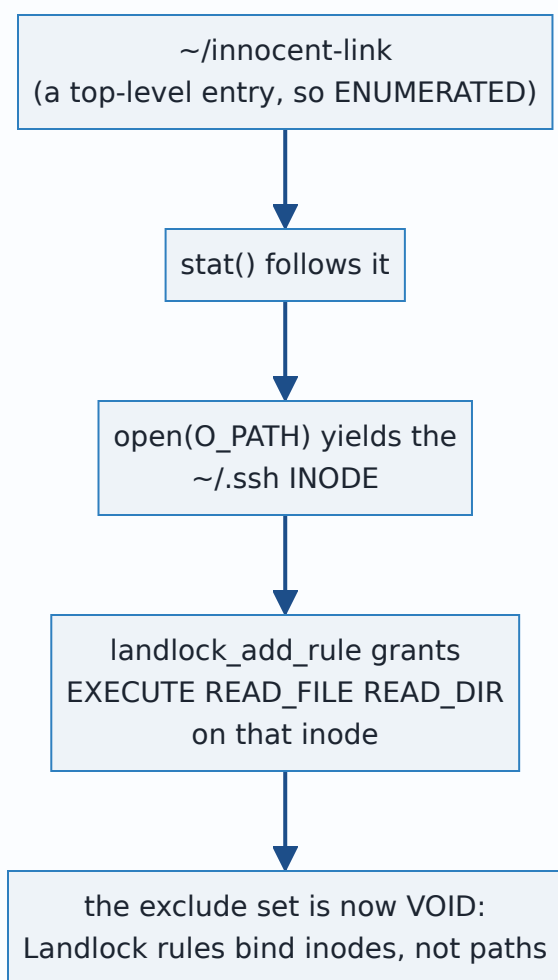


Entries are classified with `stat()`, never `lstat()` — `open(O_PATH)` follows symlinks too, so classifying by the link's own type rather than what it resolves to would grant the wrong access mode. Directories are granted `EXECUTE|READ_FILE|READ_DIR`; files, `READ_FILE` only.

## Symlinks — the enumeration **MUST** resolve them, or the exclude set is voided

An earlier revision of this ADR claimed a symlink could not route around the exclude set. **That claim was wrong, and it was wrong in the most dangerous direction.** It is corrected here rather than deleted, because the reasoning that produced it is the reasoning a future reader is most likely to repeat.

The false claim rested on a true fact: Landlock does enforce on the final resolved path. That fact is decisive for a symlink *inside* an already-granted tree, where no rule is ever created for the link itself. It is irrelevant at an **enumerated depth** — the top level of `$HOME`, and every child of `.config` — because there the enumeration creates a rule *for the entry itself*, and it classifies with `stat()`, which follows. So the rule is added on an fd opened on the **resolved inode**.



Measured on this host (Landlock ABI 8), enumerating a fixture `$HOME` containing `innocent_dir_link -> .ssh` and `.config/innocent_file_link -> .ssh/known_hosts`, with `.ssh` correctly skipped by name:

| probe                                                         | result          |
|---------------------------------------------------------------|-----------------|
| <code>/etc/hostname</code> (no rule — liveness control)       | DENIED          |
| <code>normal/plain.txt</code> (granted — control)             | OK              |
| <code>.config/innocent_file_link</code>                       | <b>readable</b> |
| <code>innocent_dir_link/known_hosts</code>                    | <b>readable</b> |
| <code>~/.ssh/known_hosts</code> <b>by its own direct path</b> | <b>readable</b> |

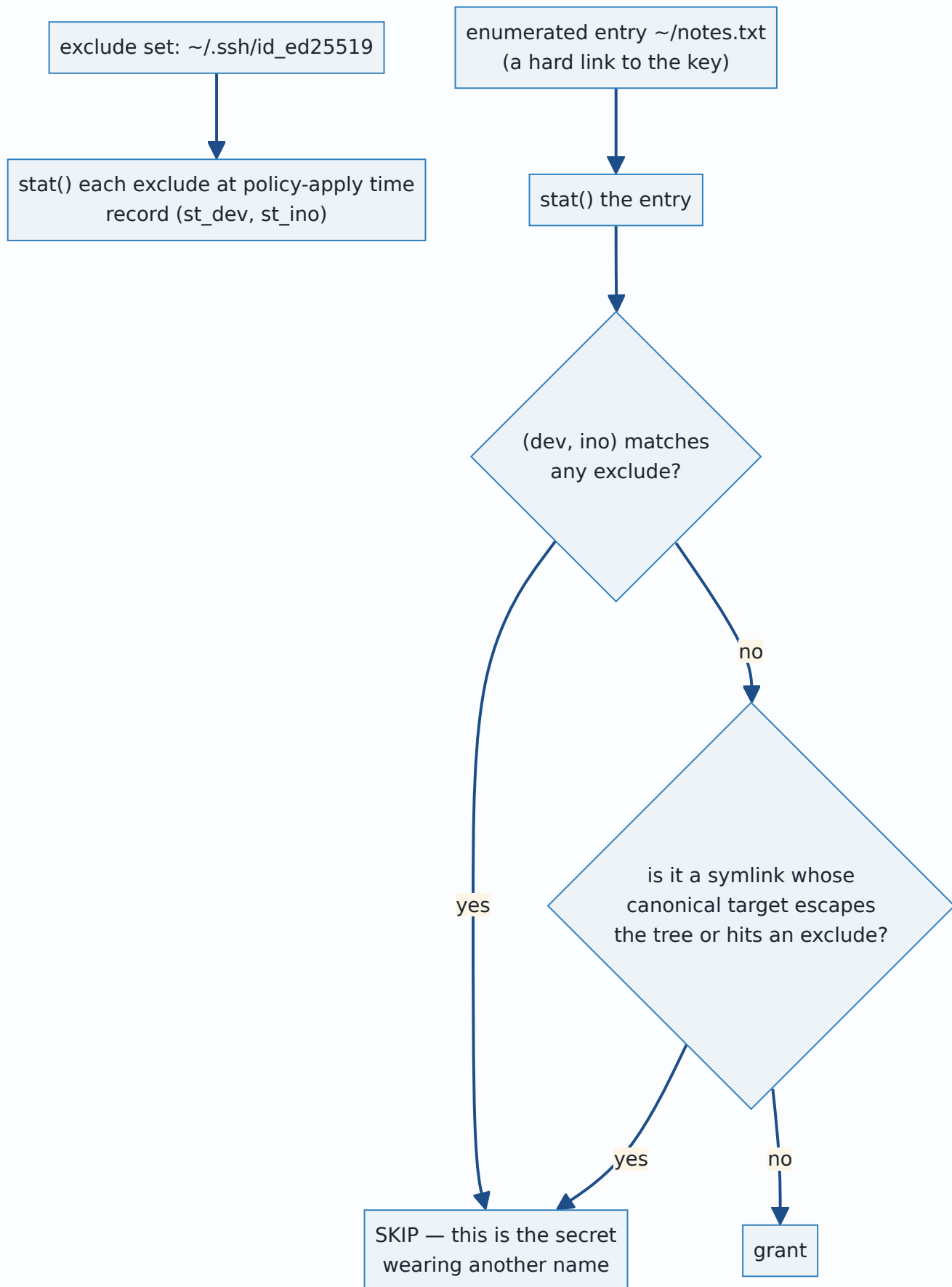
The last row is the one that matters. Granting the `.ssh` inode through an alias re-opens the *canonical* path too, because a Landlock rule is a property of the inode. A symlink at an enumerated depth does not merely bypass the exclusion — it **silently deletes it**, and every later assertion about `~/.ssh` being unreachable becomes false while the ruleset still looks correct.

This is live on any real machine: this host has seven top-level `$HOME` symlinks (`.bashrc`, `.gitconfig`, `.zshrc`, ...), all pointing into a dotfiles repository. None currently resolves into or above an excluded directory, so there is no exposure today — but the mechanism is one `ln -s` away from voiding the whole exclude set, with nothing to signal it.

### The fix is at inode identity, not at path — because hard links exist

The obvious repair is to `lstat()` each entry and, for symlinks, `canonicalize()` and skip anything whose target leaves the granted tree or touches an exclude. That is necessary and it is **not sufficient**, which an independent re-measurement established after this ADR was first written.

A **hard link has no target**. It *is* the inode, under a second name. So `canonicalize()` returns the link's own innocent-looking path, the entry passes every symlink test, and granting it re-grants the canonical secret inode exactly as the symlink case did. Measured: "following a non-secret symlink into `.ssh`, **or granting a non-secret hard link to a secret file**, re-grants the canonical secret inode." Symlink-resolution logic is structurally incapable of seeing this, because there is no link to resolve.



## Rejected: matching excluded inodes directly

The obvious repair for hard links is to record `(st_dev, st_ino)` for every inode under each exclude and skip any entry that matches. It is correct, and it was measured working — but it costs a recursive walk of every excluded tree at policy-apply time. On this host that is **10,895 entries**, dominated by `.claude` (6,568) and `.config/google-chrome` (4,006), on a path whose entire latency budget is ~9.9 ms. Rejected on cost, not on correctness.

## Accepted: `st_nlink`, which is already in the `stat()` we perform

A hard link means, by definition, that an inode carries **more than one name**. So `st_nlink > 1` on a regular file is a *necessary* condition for the entry to be a hard-link alias. Skipping every enumerated regular file with `st_nlink > 1` therefore has **no false negatives**, costs no additional syscall, and requires no walk. It is conservative — a legitimately hard-linked regular file gets skipped too — and that is the documented trade. Measured: zero top-level `$HOME` entries on this host have `st_nlink > 1`.

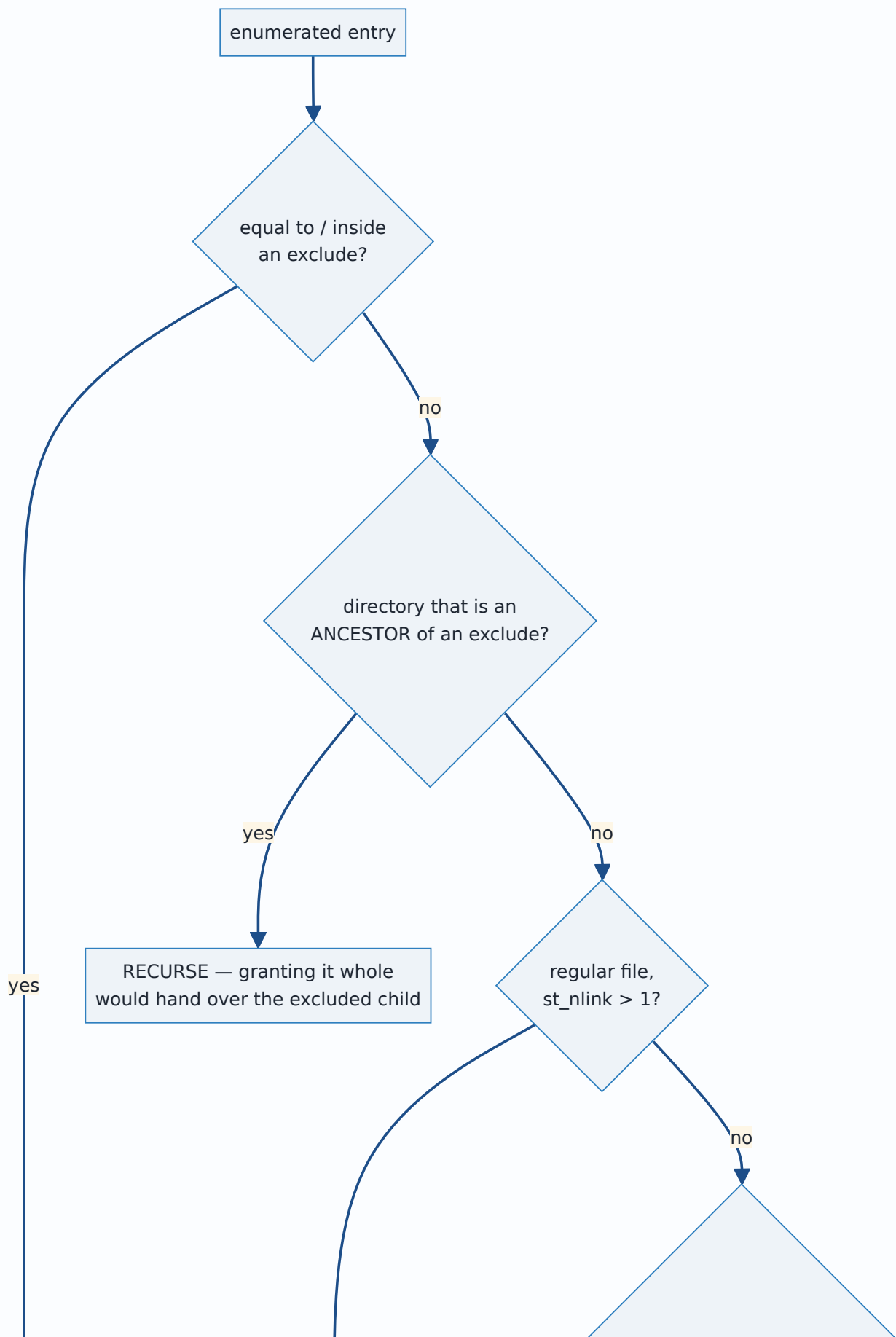
## The complete rule, and the distinction that breaks `git commit` if collapsed

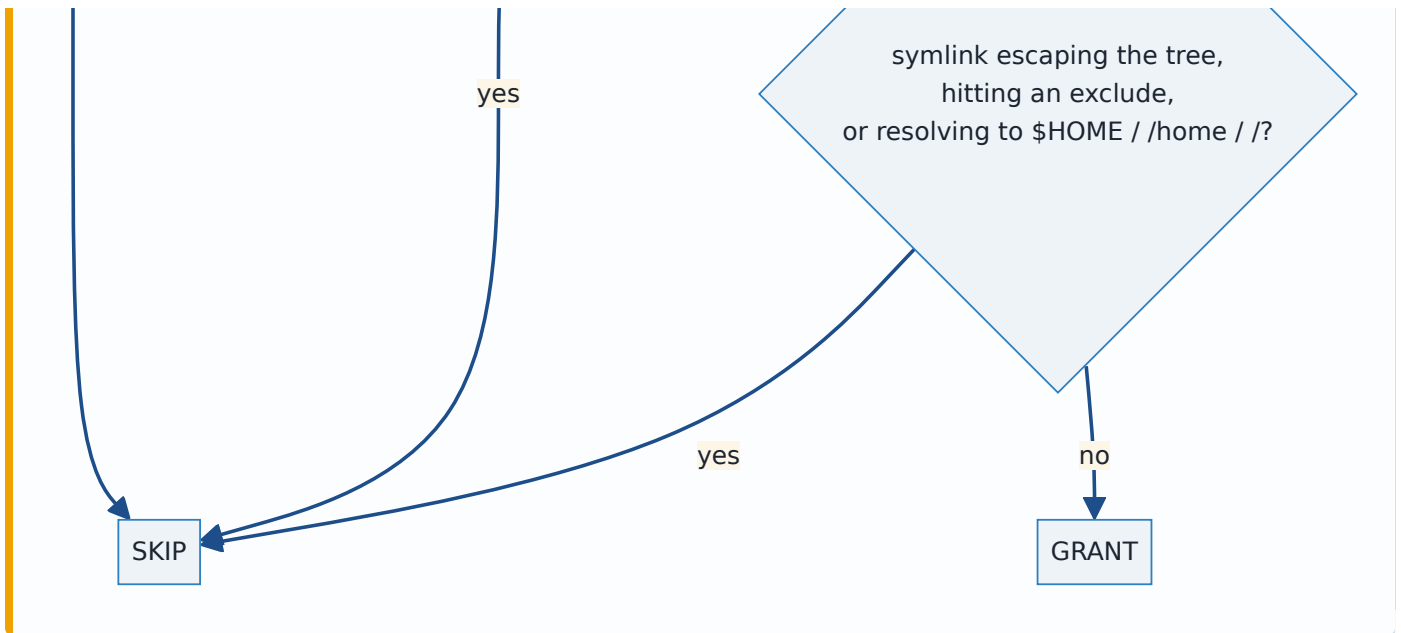
For each entry, resolve it and then:

| condition on the resolved path                                                                  | action                                        |
|-------------------------------------------------------------------------------------------------|-----------------------------------------------|
| equal to an exclude, or inside one                                                              | <b>skip</b>                                   |
| a directory that is a proper <i>ancestor</i> of an exclude                                      | <b>recurse into it — never grant it whole</b> |
| a regular file with <code>st_nlink &gt; 1</code>                                                | <b>skip</b> (hard-link guard)                 |
| a symlink whose target escapes the tree, hits an exclude, or is <code>\$HOME / /home / /</code> | <b>skip</b>                                   |
| anything else                                                                                   | grant                                         |









**Collapsing rows 1 and 2 into a single "touches an exclude → skip" test is a regression, and it is the one this project already shipped once as F-NEW-2.** `.config` is an ancestor of the exclude `.config/gh`. Treating that as *skip* drops `.config` entirely, `~/.config/git/ignore` becomes unreadable, and git fails outright:

```
warning: unable to access '/home/tom/.config/git/ignore': Permission denied
fatal: cannot use /home/tom/.config/git/ignore as an exclude file
```

Ancestor-of-exclude means **recurse**; only *equal-to* or *inside* means skip. For a symlink's canonical target the ancestor case remains disqualifying, because there the alias would grant the ancestor rather than descend into it — the same word, two different questions.

## Measured acceptance

Against the real `$HOME`, the full exclude set, and an env stripped to `PATH` and `HOME` so identity could come from nowhere else:

```
granted=172 skipped=5
DENIED-CONTROL ~/.ssh/known_hosts      DENIED
DENIED-CONTROL ~/.claude/CLAUDE.md     DENIED
GRANTED-CONTROL ~/.gitconfig (symlink)  OK
no local identity (correct)
commit_exit=0
author=tomb <thomaslane2025@gmail.com>
secret via git: denied (correct)
```

All four alias attacks — symlink-to-directory, symlink-to-file, hard link, and symlink-to- `$HOME` — were denied, **including** `~/.ssh/known_hosts` **by its own canonical path**, which is the row that distinguishes a real fix from one that only blocks the alias while leaving the exclusion void.

The test for this must assert that canonical-path row. Asserting only that the alias is denied would pass while the secret was readable by its real name.

## Where the enumeration runs

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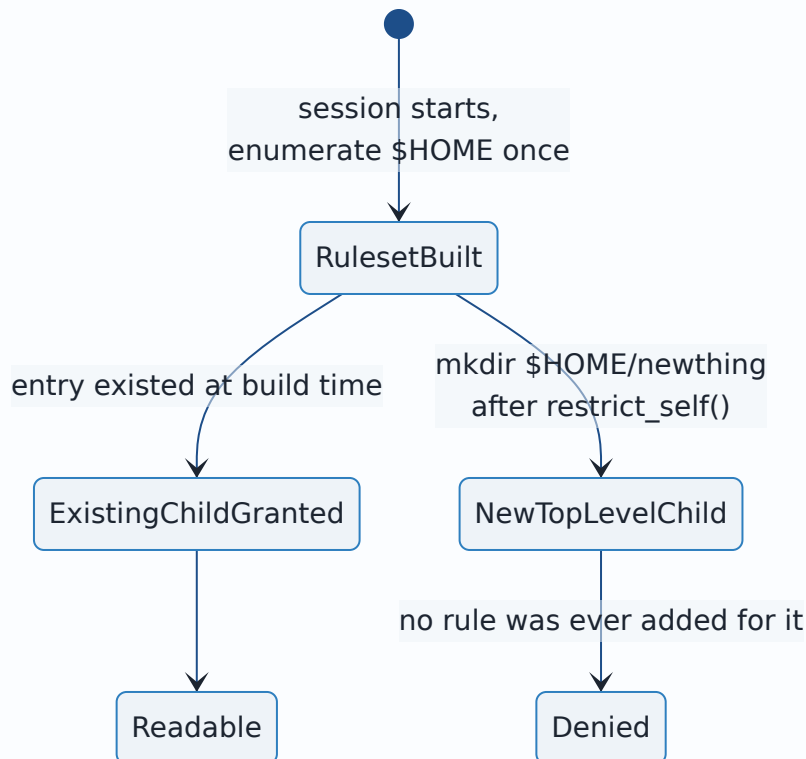
The enumeration and the `landlock_add_rule` calls it produces live in the **shim** ( `gv-sandbox` , Task 3 of the M1.13b plan — see Consequences for build status), not in the pure policy builder ( `sandbox::sandbox_argv` , `crates/git-vista-server/src/sandbox/mod.rs` ). Three reasons, all structural:

1. **sandbox\_argv stays pure.** Task 1's whole premise is a policy builder with zero I/O and zero syscalls, host-testable everywhere. Reading `$HOME`'s directory contents to build the grant set is disk I/O; doing it inside the pure function would break that premise for every existing test.
2. **The argv stays short enough to review by eye.** `$HOME` has 172 entries on this box after recursing one level into `.config`. Passing one `--ro` per entry would make the launcher argv a wall of ~170 paths instead of the handful of tree roots and exclude paths it carries today.
3. **INV-16 keeps something fixed to compare against.** The argv tripwire asserts every sandboxed launcher argv ends in a fixed reviewed shape. What travels in the argv is the short, auditable **exclude list** — nine entries — not the (much longer, and host-dependent) grant list the shim derives from it at runtime. That is exactly the property D5 Option B was chosen for: the thing a reviewer reads is the secret list, not a several-hundred-line grant dump.

## Consequences

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- **Rule count scales with \$HOME's entry count.** 172 rules for the real `$HOME` on this box today. Measured headroom: 300,000 rules were added to a single ruleset without failure — roughly **1744x** today's count — so this is a real coupling between "how much lives in `$HOME`" and "how many Landlock rules the shim builds," but it is not a constraint that bites at any size a home directory plausibly reaches.
- **A brand-new top-level \$HOME child created after the ruleset is applied is denied.** Enumerate-and-skip grants `$HOME`'s *existing* children at policy-build time; it never grants `$HOME` itself. A directory created under an already-granted parent (e.g. a new file inside `~/projects`) is visible immediately, because the parent's own grant covers it — but a wholly new top-level entry ( `~/newthing` ) has no rule at all until the next time a ruleset is built, because nothing enumerated it. This is a real, accepted limitation, not an oversight to silently work around.



- **A related correction, same mechanism, different tree.** Repositories are not cloned under `~/projects` — `resolve_clones_root` (`crates/git-vista-server/src/state.rs`) resolves to `$XDG_DATA_HOME/git-vista/clones`, falling back to `~/.local/share/git-vista/clones`. `~/.local` is granted only **read-only** as a top-level entry by `enumerate-and-skip`, so the repository's own read-write path needs a **nested, more-permissive** rule under that read-only ancestor. Measured separately, and it is the mirror image of the failed deny-rule alternative: a nested rule **does** union upward when it grants *more* than its ancestor — a write under the RW-nested subtree succeeded while a sibling without that deeper grant stayed read-only. Landlock's most-specific-rule semantics work in the permissive direction; only the restrictive direction (Alternative 2, below) is inert.
- **The enumeration itself is not yet built.** This ADR fixes the mechanism the shim must implement (Task 3 of the M1.13b plan); the pure policy-builder half — `Policy.secret_excludes`, `DEFAULT_SECRET_EXCLUDES`, and `shim_argv` emitting one `--exclude <path>` per excluded entry — is built and tested today. The shim binary that actually calls `landlock_add_rule` per enumerated entry does not exist in the tree yet. Marking this ADR "Accepted" records the *decision*, proven correct by measurement against a first-party compiled probe and an independent adversarial re-run; it does not claim the production shim runs it yet — see ADR 0025 for why that distinction is written explicitly rather than left to be inferred.

## Alternatives considered

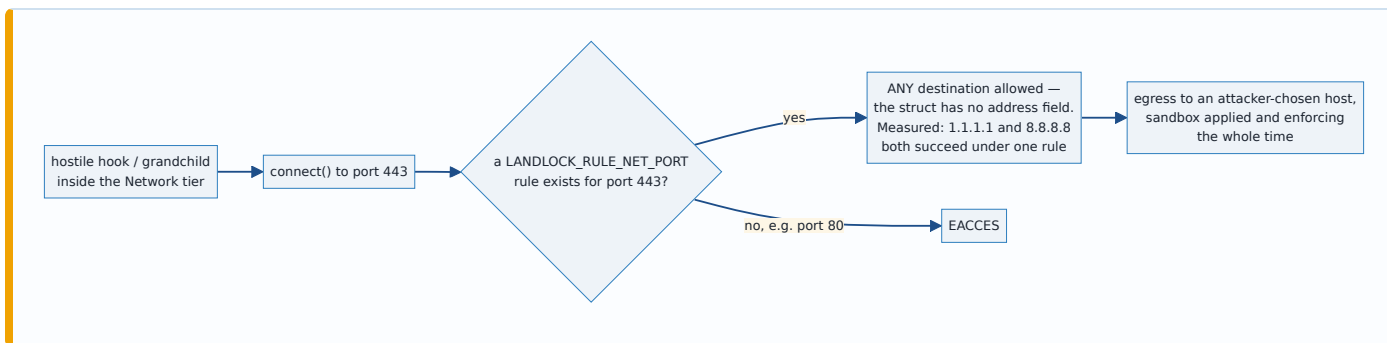
- **A path\_beneath deny rule ( allowed\_access = 0 ).** The original plan's own mechanism. Rejected: the kernel refuses the call outright ( errno 42 , ENOMSG ); shipping it would abort the shim's restrict() path via its ? operator on every host where any excluded path exists.
- **A nested lower-privilege rule** (e.g., \$HOME/.ssh granted only MAKE\_BLOCK , hoping the more specific rule wins). Rejected: measured inert — the ancestor's grant on \$HOME still applied, and the excluded file remained readable while an unrelated no-rule control correctly returned EACCES , proving the ruleset was live and the "deny" simply did nothing.
- **Blanket-deny-plus-allowlist** (grant nothing by default, allowlist every path git legitimately needs) — the shape round 4's earlier iterations tried before landing on D5 Option B. Rejected as a *general* strategy, not just for this decision: it needed three iterations to make one git commit succeed (round 4's F1 and F-NEW-2 findings), and would keep growing for every new credential helper, hook interpreter, or tool git shells out to. Enumerate- and-skip inverts this: start from "grant everything under \$HOME " (what git actually needs, unconditionally) and subtract a short, named, human-reviewable list, instead of trying to enumerate everything git might ever touch.
- **Enumerating in the pure policy builder instead of the shim.** Considered and rejected for the three structural reasons under "Where the enumeration runs" above: it would make sandbox\_argv impure, blow the argv up to roughly 170 entries instead of a handful, and remove the fixed, reviewable shape INV-16's structural assertion depends on.

## Related, not decided here: the Network tier cannot confine egress by host

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A structurally separate finding surfaced while measuring the Network tier's Landlock net rules (M1.13b's D4 Option A), and it belongs in the record even though **no decision has been made about it yet.**

struct landlock\_net\_port\_attr — the type behind LANDLOCK\_RULE\_NET\_PORT — **carries no destination-address field at all.** A single rule granting connect() on port 443 grants it to **every** destination on that port. Measured live: one port-443 rule permitted connect() to both 1.1.1.1:443 and 8.8.8.8:443 , while the same host's port 80 correctly stayed EACCES under the same ruleset — identical behavior over IPv6. This was confirmed as a *composition*, not inferred from the primitive alone: a hostile .git/hooks/pre-commit grandchild process, running under the real Network-tier ruleset alongside the real filesystem grants, read ~/.gitconfig **and** connected to 1.1.1.1:443 in the same run, while every other denied control (secret file open, connect on an ungranted port) stayed EACCES at every process level and every granted control stayed working. The filesystem boundary from this ADR held throughout that same run — a hook binary placed outside every granted tree was denied execve ( rc 126 ) — so the exploit's only opening is that .git/hooks/ must itself live inside the granted, read-write repository tree for git to run it at all.



**The consequence:** the Network tier — the one tier that must permit `push / fetch / clone` to work at all — constrains *which ports* a sandboxed process may reach, never *which hosts*. A compromised git process in that tier can exfiltrate to an arbitrary host over any granted port.

**Tom has not chosen a response.** Three options are on the table: (A) accept and document the limitation, amending `SECURITY_MODEL.md`; (B) build a real egress boundary (a proxy or an allowlisted DNS/host gate) so the tier can actually confine a compromised process to its own remote; (C) drop the net rules from the Network tier's Landlock policy entirely, since a port-only rule that cannot constrain the host it reaches may not be worth the complexity it adds. **Nothing in this record should be read as having picked one.** This ADR states the measured fact and marks the decision **PENDING** — it belongs in its own ADR once Tom rules on A, B, or C; recorded here only so a later reader does not have to re-derive the finding from scratch, or worse, assume the Network tier's Landlock net rules provide host-level confinement they structurally cannot.

## Where this is implemented

- `crates/git-vista-server/src/sandbox/mod.rs` — `DEFAULT_SECRET_EXCLUDES`, `Policy.secret_excludes`, and the doc comment recording both measured kernel failures (`ENOMSG`, the inert nested rule) and the enumerate-and-skip replacement; `shim_argv` emitting one `--exclude <path>` per excluded entry.
- `crates/git-vista-server/src/sandbox/argv.rs` — `the_argv_never_names_a_deny_rule` (the tripwire: if `--deny` ever reappears in a launcher argv, the broken mechanism has been reinvented and the secret set is silently readable again), `grants_and_excludes_are_passed_through_as_separate_argv_entries`.
- **Not yet built:** the `gv-sandbox` shim binary itself (Task 3 of the M1.13b plan) — the process that actually walks `$HOME`'s entries and calls `landlock_add_rule` per the algorithm this ADR fixes. This ADR records the mechanism the shim must implement, ahead of that code existing, precisely so the mechanism is not re-derived (or re-broken) by whoever writes it.
- Measurement evidence: first-party compiled probes plus an independent adversarial re-run, referenced from `handoff.md` (repo root, gitignored — not the durable record; this ADR is).

**Signed:** thomas2025 · 2026-07-29T03:17:07-04:00